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Summary Computer - Extensive Summary Of The First 8 Chapters Of The Book

History of Science (Vrije Universiteit Amsterdam)

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SUMMARY History of Science

Comprising chapters 1-8

Book

Computer: A History of the Information Machine (third edition) by Campbell-Kelly, M., Aspray, W., Ensmenger, N. & Yost, J. R.

Vrije Universiteit 2015-2016

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Introduction

The old technologies had three shortcomings: 1) they were too slow in doing their calculations, 2) they required human intervention in the course of a computation, and 3) many of the most advanced calculating systems were special-purpose rather than general-purpose devices. Decades of continuous innovation have been of two types:

- One is the improvement in components, leading to faster processing speed, improved price/performance, better reliability, etcetera. (agent of change: computer manufacturers);
- The second type of innovation was in the mode of operation, high-level programming languages, real-time computing, time-sharing, networking, and graphically oriented human-computer interfaces. (agent of change: academic sector, financed by government).

When the mainframe became more refined, unfortunately, merely the privileged few had access to the computer. All of this changed with the development of the personal computer and the growth of the Internet. Computing became available to the masses.

PART I BEFORE THE COMPUTER

1 When computers were people

The electronic computer can be said to combine the roles of the human computer and the human clerk. The first attempt to organise information processing on a large scale using human computers was for the production of mathematical tables.

For a maritime nation such as Great Britain, and later the United States, the timely production of reliable navigation tables free of error was of major economic importance. This seaman's bible was called the *Nautical Almanac* (1766), an annual set of navigational tables. The calculations were performed twice, independently, by two computers and checked by a third "comparator."

De Prony organised his table-making team for the *Tables du Cadastre* according to the famous pin-making factory example: 1) those that decided on the mathematical formulas to be used in the calculations, 2) those that organised the computations and compiled the results ready for printing, and 3) those that did the actual computation.

<u>Babbage and his Engines</u>
Charles Babbage (mathematician and economist, founder of the Analytical Society) became interested in the problem of table making and the elimination of errors in tables. His job was to supervise and check the accuracy of the calculations. Inspired by De Prony's way of organisational technology, he produced the Difference Engine (with financial help of the government), which did some calculations and printed them. This, while office machinery were a scientific novelty at that time. Unfortunately, the engineering was more complicated than he had foreseen. He was paying the price of a first-mover. Although he never finished the machine, he did engineer a prototype which showed beyond any question the feasibility of his concept. Then, he wanted to make the Analytical Engine, which would be capable of performing *any* calculation that a human could specify for it. But he did not receive another penny from the government because the Difference Engine "failed".

<u>Clearing houses</u>
The Bankers' Clearing House was an organisation that processed the rapidly increasing number of checks being used in commerce. Clerks from all different banks had to meet at to the same clearing room at the same time. There they performed all the exchanging of checks and cash in one "clearing room." This obviously saved a lot of walking time and avoided the danger of robbery. It also brought to light that if two banks had checks drawn on each other, the amount of cash needed for settlement was simply the difference between the two amounts owed, which was usually far less than the total amount of all the checks.

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Telegrams

It began as a solution to a communications problem in the early rail system. Telegrams controlled that, now, a train could not enter a single-track section until two human operators agreed that it was safe to do so. Telegrams were expensive—one shilling for a twenty-word message compared with as much as you could want to write in a letter for one penny—but very fast. A telegram would speed across the country in an instant and arrive at its final destination in as little as an hour, hand-delivered by a telegraph boy. Although messages were sent by electricity, the telegraph still needed a large (female) clerical labor force to operate the machines that transmitted the messages. Because each system operated independently, it was difficult for a telegram originating in one network to make use of another. In 1870, the British government stepped in to integrate the systems into a national telegraph network. Once this was done, telegraph usage simply exploded and costs fell to sixpence for a dozen words. Telegraphists needed to have a good knowledge of national geography. In 1874 a central hub, the Central Telegraph Office, was established with a direct connection to "every town of importance in the United Kingdom."

Herman Hollerith and the 1890 census

Compared to Europe, the United States was a latecomer to large-scale data processing. That's because it lagged twenty or thirty years behind Europe in its economic development. But this delay enabled them to take full advantage of the newly emerging office technologies.

The Bureau of the Census made a count of the entire population of the United States every ten years. The data-processing method in use then was known as the "tally system." Because the population increased so fast whereby the census took almost seven years to process, there was a strong need for mechanisation. Hollerith is regarded as one of the seminal nineteenth-century figures in the development of information processing. He had very practical (not at all polymathic) and had a strong entrepreneurial flair. Hollerith's key idea was to record the census return for each individual as a pattern of holes on punched paper tape or a set of punched cards, similar to the way music was recorded on a string of punched cards on fairground organettes of the period. It would then be possible to use a machine to automatically count the holes and produce the tabulations.

After he commercialised this invention by establishing the Tabulating Machine Company in 1896, he laid the foundations of IBM. The great advantage of the Hollerith system was that once the cards had been punched, all the sorting and counting would be handled mechanically. Because of this, it was over ten times as fast as his rivals. Hence, the Hollerith Electric Tabulating System was adopted for the 1890 census. The Hollerith machines had opened up a whole new way of mechanised information processing.

America's love affair with office machinery

Example office machines: typewriters, record-keeping systems, and adding machines. The Hollerith system was not in any sense typical of this information technology. Around 1890-1900, office equipment, in both its most advanced and its least sophisticated forms, was almost entirely an American phenomenon. Its like would not be seen in Europe until after World War I. Redesigning the office system to make use of typewriters, adding machines, or modern card index systems was never cost-effective. But because America was late with implementing them, they did not endure this hassle.

The fact is that America was gadget-happy and was caught up by the glamour of the mechanical office. American firms often bought office appliances simply because they were perceived as modern. Therefore, the United States was the first country in the world to adopt office machinery on a large scale.

2 The mechanical office

Examples of top office-machine suppliers: Remington Rand, National Cash Register (NCR), the Burroughs Adding Machine Company, and IBM.

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Today, we use computers in the office for three main tasks: document preparation, information storage, and financial analysis and accounting.

The typewriter

Remington introduced the typewriter around 1870. On the very first machines, the letters of the keyboard had been arranged in alphabetical order, and the major cause of the jamming was the proximity of commonly occurring letter pairs (such as D and E, or S and T). The easiest way around this jamming problem was to arrange the letters in the type-basket so that they were less likely to collide. The result was the QWERTY keyboard layout that is still with us. Because of the investment in the training of typists, and the natural reluctance of QWERTY-familiar people to change, there has never been a right moment to switch to a more convenient keyboard layout. And maybe there never will be.

Mark Twain, an early enthusiast, wrote the most famous testimonials for Remington about the annoying fact that if he wrote his acquaintance a letter with the typewriter, he would receive a letter back with HOW DID YOU DO THAT?! So, he never wrote a letter with that thing again.

Remington introduced the shift key in 1878. By 1880, Remington virtually had a monopoly on the typewriter business. One of the crucial business problems that Remington addressed at an early date was sales and after-sales service. So some competitors joined the field. Typewriters became the most widely used business machines, also due to its low price (just $75). The training of workers to use them was very important, namely, it took several months of intense practice to acquire the skill. Office work was mostly female.

The typewriter was a progenitor of the computer industry, because it pioneered three key features of the office-machine industry: 1) the perfection of the product and low-cost manufacture, 2) a sales organisation to sell the product, and 3) a training organisation to enable workers to use the technology.

The Rands

The Rands were the inventor-entrepreneur proprietors of the Rand Kardex Company, the world's leading supplier of record-keeping systems. One of their innovations was the vertical filing system, but it has become so commonplace that its significance is rarely noted. They also invented and patented a "visible" index system of dividers, coloured signal strips, and tabs, enabling documents to be held in vertical files and located through an index system three or four times faster than previously. Later, Remington merged with this company —> Remington Rand.

The first adding machines

Financial organisations needed a written record of numbers as they were entered into the adding machine so that they would have a permanent record of their financial transactions. The machine (the Comptometer or the Burroughs Adding Machine) was key-driven, like the typewriter, and not dialled. The keys were arranged in ten columns, labeled 1 through 9 in each column. The skill to use such a machine was not easily acquired. Hence the company established Comptometer training schools. As with the typewriter, sales were slow at first, but by 1900 the firm was selling over a thousand machines a year. A major impetus to development of the US adding-machine industry occurred with the introduction of a new tax law that required more complex rates etcetera.

The Burroughs Adding Machine was the first to fulfil the need of a printing component. Burroughs also was the only company of all American adding-machine firms that made a successful transition into the computer age. This was because:

- of the intangible know-how of incorporating the machines into an existing business organisation;
- they did not cling to a single product but, instead, gradually expanded its range in response to user demand;
- of the knowledge of business accounting systems became deeply embedded in the sales culture of Burroughs.

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The National Cash Register (NCR) Company
NCR largely pioneered on innovative forms of selling such as the systematic analysis of customer requirements, the provision of after-sales support, and user training. Patterson, the founder of NRC, had a significant amount of idiosyncrasies, but nevertheless possessed an instinct for organising the cash register business, which was so successful that NCR's "business practices and marketing methods became standard at IBM and in other office appliance firms by the 1920s." For example, he understood the product needed constant improvement, this is from which the first formal R&D branch was developed.

NCR's greatest legacy to the computer age was the way it shaped the marketing of business machines and established virtually all of the key sales practices of the industry. It would be wrong to credit Patterson with inventing all of these sales techniques. Certainly he invented some basic ideas, but mostly he embellished and formalised existing practices. Some examples: the concept of the sales quota, which was a stick-and-carrot sales incentive; *The Hustler* (booklet that showed prospective users how to organise retail accounting systems and gave inspirational encouragement); and sales training.

Thomas Watson and the founding of IBM
An early graduate of the NCR sales school was Thomas J. Watson. Attending the school changed his life. Watson became an outstanding salesman and had a lot of promotions in his career. He is also responsible for the famous and simple slogan, T-H-I-N-K. After Watson was fired by Patterson, he became the president of C-T-R. He completely transformed the culture and prospects of C-T-R, turning it into the brightest star of the office-machine industry. Hollerith was very wealthy with TMC (specialised in punched-card machinery), but because his health was declining, he decided to merge with C-T-R, and ten years later Watson renamed the company to IBM.

IBM survived the Great Depression and became a business-machine legend, which made Watson the most influential businessman. There were three reasons for the ultimate success of IBM:
- The development of a sales organisation based on that of NCR;
- The "rent-and-refill" nature of the punched-card machine business (this virtually made it recession proof);
- Technical innovation.

IBM's financial stability can also be attributed to its punched-card sales. The cards cost only a fraction of the dollar per thousand for which they were sold. IBM, however, found itself secure in the market of tabulator cards for its machines, because the cards had to be made with great accuracy on a special paper stock that was never successfully imitated.

3 Babbage's dream comes true

In October 1946 the computing pioneer Leslie John Comrie reminded British readers in a science news journal of their government's failure to support the construction of Babbage's calculating engine a century earlier: "It is not too much to say that it cost Britain the leading place in the art of mechanical computing".

Babbage's Analytical Engine
As important as the Difference Engine had been, it was fundamentally a limited conception in that all it could do was produce mathematical tables. By contrast, the Analytical Engine would be capable of any mathematical computation. The idea of the Analytical Engine came to Babbage when he was considering how to eliminate human intervention in the Difference Engine by feeding back the results of a computation.

The most significant concept in the Analytical Engine was the separation of arithmetic computation from the storage of numbers. For about two years Babbage struggled with the problem of organising the calculation—the process we now call programming, but for which Babbage had no word. There was no question at this time of his attempting construction of the machine without

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government finance, so it was largely a paper exercise—eventually amounting to several thousand manuscript pages. During this period of intense creative activity, Babbage essentially lost sight of the objective of his engines. Ada translated an account of the Analytical Engine of some mathematician and even added lengthy notes of her own that almost quadrupled the length of the original. Her *Sketch of the Analytical Engine* was the only detailed account published in Babbage's lifetime. Therefore, Babbage described her as his "dear and much admired Interpreter."

Babbage's view, however, was that once he had hit upon the Analytical Engine, it was a pointless distraction to press ahead with the inferior Difference Engine merely to demonstrate concrete results. The Swedish men however, learned from the existence of the machines and they completed a full-scale engine that rivalled Babbage's own. He was gratified to see the engine win the Gold Medal in its class in 1852. The primary cause of Babbage's failure was that he was pioneering in the 1820s and 1830s a digital approach to computing some fifty years before mechanical technologies had advanced sufficiently to make this relatively straightforward.

The tide predictor and other analog computing models

The concept behind this mode of computing is that instead of computing with numbers, one builds a physical model, or analog, of the system to be investigated. When a system could not be readily investigated mathematically, the best way forward was to build a model. The most important analog computing technology of the nineteenth century, from an economic perspective, was the mechanical tide predictor, invented by Lord Kelvin. The overriding problem with analog computing technologies was that a specific problem required a specific machine.

The differential analyser of Bush however, was not a general-purpose computer in the modern sense, but it did address such a wide range of problems in science and engineering that it was far and away the single most important computing machine developed between the wars. It fitted with those problems for which it was possible to construct a mechanical or electrical model of the physical system.

A weather-forecast factory

For other problems, (e.g. table-making, weather-forecasting) it was necessary to work with the numbers themselves. This was the digital approach. Lewis Fry Richardson pioneered numerical meteorology—the application of numerical techniques to weather forecasting. Richardson became convinced that forecasting tomorrow's weather on the basis of past experience—in the belief that history would repeat itself—was fundamentally unscientific. He believed that the only reliable predictor of the weather would be the mathematical analysis of the existing weather system. Weather prediction by Richardson's method involved an enormous amount of computation. 32 human computers were needed to keep pace with the weather of one region. He described his weather-forecasting factory fantasy which was well received, but never taken seriously.

The scientific computer service

Leslie John Comrie was elected director of the computing section of the British Astronomical Association, where he organised a team of twenty-four human computers to produce astronomical tables. Comrie's great insight was to realise that one did not need special-purpose machines such as differential analysers; he thought computing was primarily a question of organisation. Moreover, his major achievement at the Nautical Almanac Office was to bring punched-card machines—and therefore, indirectly, IBM—into the world of scientific computing.

Comrie set up a business as a commercial operation, the Scientific Computing Service Limited. It was the world's first for-profit calculating agency. The timing was perfect due to the arrival of World War II.

The Harvard Mark I

The fact that there were so many human-computing organisations in existence is indicative of the demand for computing in the years up to World War II and during the war itself. The Harvard Mark I

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was the best known, and the earliest machine that was typically used for table making, ballistics calculations, and code-breaking.

Bush's differential analyser at nearby MIT were suitable for ordinary differential equations, they could not generally solve nonlinear equations. Whereby Aiken, the chief beneficiary of IBM's computing largess, consciously saw himself as Babbage's twentieth-century successor, after he had heard of him and his Engines. He proposed a calculator at IBM and essentially merely fulfilled the role of a consultant. The five-ton calculator was a monumental piece of engineering. The machine could store just seventy-two numbers and was capable of three additions or subtractions a second. The significance of the Mark I, however, lay not in its speed but in the fact that it was the first fully automatic machine to be completed. Unfortunately, the calculator was incapable of making what we now call a "conditional branch"—that is, changing the progress of a program according to the results of an earlier computation. This made complex programs physically very long (Babbage had already arrived at the concept of this issue). At the ceremony of the Mark I, Aiken's arrogance overcame his judgment to such a degree that his reputation never fully recovered.

The Harvard Mark I, however, was a fertile training ground for early computer pioneers. But above all the Harvard Mark I was an icon of the computer age—the first fully automatic computer to come into operation in 1943. The machine also had an emotional appeal as a final vindication of Babbage's life.

Theoretical developments and Alan Turing

There were theoretical developments in mathematical logic that would later have a major impact on the progress of computer science as an academic discipline. The *Entscheidungsproblem,* asked: "Did there exist a definite method or process by which all mathematical questions could be decided?" That is where Turing came up with a "universal" computer that was maybe more a thought experiment than a realistic computer. He describes the Turing Machine in his classic paper. The machine embodied all the logical capabilities of a modern computer.

There he played a vital part in using computing machinery to break Nazi codes (it is believed that his work shortened the war in Europe by many months). Sadly, Turing died due to indirect effects of his deviant sexual preferences. It is likely that Turing's work significantly influenced von Neumann and, hence, the invention of the modern electronic computer.

PART II CREATING THE COMPUTER

4 Inventing the computer

Bush established the National Defense Research Committee (NDRC), which would "coordinate, supervise, and conduct scientific research of existing projects." It consisted of twelve leading members of the scientific community who had the authority and confidence to get business done.

The Moore School

The Ballistics Research Laboratory (BRL) calculated the trajectories of weapons fired through the air or water. What brought the BRL and the Moore School of Electrical Engineering together was the fact that they both had a Bush Differential Analyzer. In the months preceding World War II, the Moore School was placed on a war footing: the main training activity was the Engineering, Science, Management, War Training (ESMWT) program, which was an intensive ten-week course designed to train physicists and mathematicians for technical posts. All this computing activity was needed to produce "firing tables" (pocket-size booklet) for newly developed artillery, and for older equipment being used in new theatres of war, just a rule-of-thumb was simply not sufficient to find the range. The lack of an effective calculating technology was a major bottleneck to the effective deployment of the multitude of newly developed weapons (calculations took way too long).

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The Atanasoff-Berry Computer (ABC)

In the summer of 1942 John Mauchly proposed the construction of an electronic computer to eliminate this bottleneck. The project was actually begun in 1937 by John Vincent Atanasoff (described by his biographer as "forgotten father of the computer"), a professor of mathematics and physics. He enlisted the help of a graduate student, Clifford Berry. They developed several ideas that were later rediscovered in connection with electronic computers, such as binary arithmetic and electronic switching elements. They shared their ideas with Mauchly. He subsequently stated that he took "no ideas whatsoever" from Atanasoff's work in his own project. The extent to which he drew on Atanasoff's ideas remains unknown, and the evidence is massive and conflicting. At the very least, we can infer that Mauchly saw the potential significance of the ABC and that this may have led him to propose a similar, electronic solution to the BRL's computing problems.

John Presper Eckert, a twenty-two-year-old research associate who had only recently completed his master's degree, was "undoubtedly the best electronic engineer in the Moore School." Mauchly and him frequently exchanged ideas about electronic computing—Mauchly invariably in the role of a conceptualiser and Eckert in that of a feet-on-the-ground engineer. Eckert was associated with the delay-line storage system. This achieved storage by feeding back the signal a millisecond later. They were also working on the use of high speed vacuum tube devices for calculating.

ENIAC and EDVAC: the stored-program concept

"Project PX" consisted of constructing this calculating machine, known as the ENIAC (Electronic Numerical Integrator and Computer). The ENIAC was many times more complex than any previous electronic system. Although the ENIAC was Eckert's (the "consummate engineer") and Mauchly's (the "visionary") project, the achievement can much be attributed to Eckert. The ENIAC was not without its critics. Indeed, within the NDRC the attitude was "at best unenthusiastic and at worst hostile." This was mostly due to the life expectancy of a tube: by a naive calculation, that there would be a tube failure every ten minutes. Perhaps the most serious problem was the time it would take to reprogram the machine when one problem was completed and another was to be run.

Goldstine, a member of BRL, introduced John von Neumann to the project. He evaluated the machine and came to the conclusion that there were three major shortcomings of the ENIAC: too little storage, too many tubes, and too lengthy reprogramming. Von Neumann's arrival gave the group the confidence to submit a proposal to the BRL to develop a new, post-ENIAC machine. It helped that von Neumann was able to attend the BRL board meetings where the proposal was discussed. Funding was provided for "Project PY": the EDVAC, for Electronic Discrete Variable Automatic Computer.

Eckert made a great new technological invention that would solve the storage problem: by connecting the output of the mercury delay line (using appropriate electronics) to the input, the 1,000 bits of information would be trapped indefinitely inside the delay line, providing permanent read/write storage. Von Neumann shortly came by and so the stored-program concept was born: *the computer's storage device would be used to hold both the instructions of a program and the numbers on which it operated.* It was simple—once one had thought of it. It would enable a program to treat its own instructions as data. In subsequent periods of the project, there opened a schism in the group between the technologists (Eckert and Mauchly) on the one side and the logicians (von Neumann, Goldstine, and Burks) on the other, which would lead to serious disagreements later on. Namely, von Neumann decided to write down in a report how the project was evolving but did not put the names of all the other co-inventors on it and subsequently distributed the report all over the world. Moreover, as a mathematician rather than an engineer, he believed that it was critically important to distinguish the logical design from the engineering implementation—probably a wise decision given the immaturity of computer engineering and the fact that it was undesirable to lock in too early on an untried technology. He just wished to see the idea move into the public domain as rapidly as possible, but Eckert and Mauchly saw the computer as an entrepreneurial business opportunity. In short, his academic interests conflicted with Eckert's

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commercial interests. So they split up. Then, the NDRC came around and suddenly computers were in the air.

Travis, the new research directer at the Moore School requested Eckert and Mauchly to assign future patent rights to the university, which they absolutely refused and resigned. They joined forces and set up in business as the Electronic Control Company to build computers. In the meantime there was intense interest in the ENIAC from the outside world, scientifically as well as publicly. It performed operations a thousand times faster than the Harvard Mark I.

Dean Pender ensured that the knowledge of stored-program computing was effectively transmitted to the outside world by organising The Moore School Lectures — an invitation-only summer school. The promising young scientists attending the lectures could only bring their personal notes home about the ENIAC, and limited information was given on EDVAC's design (because Project PY was still classified, however, the design was classically simple for them to find it useful).

Maurice Wilkes and EDSAC

Britain was the only country other than the United States not so devastated by the war that it could establish a serious computer research program. Because of their limited budgets, their projects were forced into keeping things simple and thus avoid the engineering setbacks common in the United States.

Newman, Williams (later Sir Frederic) and an assistant developed a simple memory system based around a commercially available cathode-ray tube. The "Manchester Baby Machine" was born, which established the feasibility of the stored-program computer.

Maurice Wilkes was a mathematical physicist that after his war service (where he also learnt engineering) reestablished the computing laboratory at Cambridge. In 1946, he visited Comrie which lent him "Draft report on the EDVAC." He couldn't copy it so he read it in one night and had no doubt that the principles of computer development described in the draft was exactly how it actually would go: the stored program (for numbers and instructions), the serial execution of instructions, and the use of binary switching circuits for computation and control.

Then Wilkes was invited by Dean Pender to attend the Moore School, which he arrived 6 weeks late to. But he was confident that the first part of the course comprised mathematics so he didn't miss much. Indeed, in this stage, the design of the EDVAC was no more than a paper design study, hence, the physical implementation was omitted. After the course, Wilkes began to sketch the design of a "computer of modest dimensions very much along the lines of the EDVAC Proposal." Namely, he was more interested in *having* a computer than advancing computer engineering technology. For this reason he chose to use delay-line storage rather than the more difficult, yet fast, CRT-based storage. Building it was only little troublesome, e.g. the supply of electronic components in postwar Britain was very uncertain and required careful planning and advanced ordering. But he was lucky to receive surplus stocks from the Ministry of Supply. In 1949 the result of this was the EDSAC: Electronic Delay Storage Automatic Calculator. Eventually, it was a big machine but it only had one-sixth the tube count of the ENIAC and was correspondingly smaller. The world's first practical stored-program computer (where programs were punched on telegraph tape and results were printed on a teletype page printer) had come to life, and with it the dawn of the computer age.

5 The computer becomes a business machine

In the 1950s the computer was reconstructed (mainly by computer manufacturers and business users) to be an electronic data-processing machine rather than a mathematical instrument. When IBM recognised this change, they adjusted their R&D, manufacturing, and sales organisations, and used its traditional business strengths to dominate the industry within a period of five years. US firms had the biggest market for computers.

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Britain was actually the first to have a computer on the market (Ferranti Mark I, Manchester University), but due to a lack of enthusiasm of using them in their old-fashioned businesses, Britain's computer industry was fighting for survival. As for the rest of continental European countries, they were so ravaged by World War II that they would not be capable of entering the computer race until the 1960s.

Three types of firms that entered the computer industry:

- Electronics and control equipment manufacturers (e.g. RCA, General Electric, Honeywell), for which computer (they coined the term *mainframe* for this) engineering was a natural step;
- Office-machine companies (e.g. IBM, Remington Rand, Burroughs, NCR), which entered the race when Eckert and Mauchly produced the UNIVAC computer in 1951, which was later than the first firm type;
- Entrepreneurial start-ups (e.g. Eckert and Mauchly's Electronic Control Company, Engineering Research Associates (ERA) and other) had a narrow window of opportunity from 1948 to 1953 to join the computer business at a comparatively love cost. These were soon taken over by office-machine companies, but most simply faded away, unable to survive in the intensely competitive environment.

"More than optimistic": UNIVAC and BINAC

When Eckert and Mauchly established their Electronic Control Company in 1946, they hired a noteworthy number of employees and they were almost unique in seeing the potential for computers in business data processing, as opposed to science and engineering calculations. Regarding their estimations for developing their first machine (an UNIVAC: UNIVersal Automatic Computer) Eckert and Mauchly were so optimistic that they were naive. They funded their first machine by letting their first order (by the Census Bureau) be paid in advance, but this was probably $700,000 short from the actual costs of the development. They hoped to recover their losses in subsequent sales. But the engineering was very hard because some basic technology needed for the machine had not even been established yet. In addition, it was one thing to develop a system for laboratory use but quite another to produce a reliable system for fuss-free use in a commercial environment. The most ambitious and revolutionary feature of the UNIVAC was the use of magnetic-tape storage to replace the millions of punched cards, unfortunately it was also the least successful aspect of the project.

Because of the financial situation of Electronic Control Company, they had no other choice than to detract momentum of the UNIVAC when an order for BINAC: BINary Automatic Computer (a scientific machine, not a data processor) was placed by Northrop Aircraft Corporation. Otherwise they had to go out of business. However, the search for more UNIVAC orders continued because it remained the long-term ambition of the company. Then in 1948 two orders came in for the UNIVAC for $150,000, which was ludicrously unrealistic of course, but Eckert and Mauchly, as always, wished away the problems. For a little while there was hope because Henry Strauss (remarkable inventor) wanted to invest in their company by shareholding 40 percent of it. However, this commitment died when he died a few weeks later and they were thrown in immediate debt again, while the future finally seemed bright again now that they also got to finish the BINAC.

IBM: Evolution, not revolution

Meanwhile, Eckert and Mauchly's future main competitor, IBM, was beginning to respond to the the computer. The uncertainty of the market however somewhat delayed their laboratory development projects to be turned into projects. Postwar, IBM was trying to respond to three critical business challenges:

- Product obsolescence (most critical one because R&D had war-specific priorities rather than improving their current products. Therefore, the latter was the top postwar priority);
- Electronics, which had greatly accelerated by the war. The race was now to incorporate them in the existing products. That is why IBM institutionalised the slogan "evolution not revolution," meaning that they would only improve their current products and not change them in any way;
- The stored-program computer (this was only important later because at first there was no obvious commercial importance!).

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Thomas Watson Sr. was among the first to see the future potential of electronics. But finding an expert was not successful because they were all committed to the war effort at that time. Not much more happened for a few years, except for the unpleasant incident when the Mark I calculators was inaugurated at Harvard University in 1944 and Howard Aiken refused to acknowledge IBM's role in inventing and building the machine. This resulted in IBM creating the more powerful electronic calculator (SSEC: Selective Sequence Electronic Calculator) with help of Wallace Eckert (not related to Presper Eckert). Besides sticking it to Aiken, the aim of developing this machine was to test for new ideas and devices, not to develop this machine into a saleable product.

In the meantime when there were no commercially available computers, IBM incorporated electronics into the company's existing products, resulting in speed improvements and much more sales. For example Model 604 calculating punch (derived from Model 601 multiplying punch), its performance was unmatched by any calculator in the market for some time and it provided the hear of a powerful computing setup known as the CPC: Card Programmed Calculator which was developed in cooperation with Northrop Aircraft. It comprised Model 603 hooked up to a storage unit and punched-card equipment. The result was a highly effective calculating setup capable of about one thousand operations per second. When other aerospace companies heard of it, IBM had received dozen requests for a similar setup. This showed that there was a significant market for scientific computing equipment. IBM decided to use the more powerful 604 for these orders. Even when computers became commercially available, the low cost and superior reliability of the CPC remained the most cost-effective computing system available. Commentators often failed to realise the importance of the CPC because it was called a "calculator" instead of a "computer" (same for the Harvard Mark I and the SSEC). Watson insisted on this terminology however because "computer" previously always referred to a human being, and he didn't want to imply technological unemployment. But as far as effective scientific-computing products were concerned, IBM had the lead from the very beginning.

The SSEC did a great deal to enhance IBM's reputation as a leader in computer technology. There was, however, no prospect that the SSEC could become a product. It cost far too much ($950,000) and was technically obsolete as soon as it was built. Its real importance to IBM, apart from publicity and image, was to create a cadre of experienced computer engineers. Other full-scale computers under development in 1949 were the Magnetic Drum Calculator (MDC) and the Tape Processing Machine (TPM). The former was a very economical product for giving true stored-program computing, the key behind it was the use of a magnetic drum for the main memory instead of delay lines or tubes, but they were quite slow too. the MDC did not reach the market soon due to conservatism in IBM's marketing, the MDC would be too expensive to compete with ordinary punched-card machines but too slow to compete with the UNIVAC. The TPC competed with the UNIVAC by being a large, tape-based data-processing computer but did not reach the market soon either because of the Korean War and a strategic decision made by Watson's son, Watson Jr. who started to lead IBM in 1956. He committed the company to developing the Defense Calculator at the expense of the TPM. Thus, while accelerating IBM's entry into scientific computers, they dashed its chances of an early entry into the data-processing market. It was a business error, one that allowed Eckert and Mauchly to capture the early market for data-processing computers with their UNIVAC.

<u>UNIVAC comes to life</u>
Eckert and Mauchly desperately needed money to sustain the momentum of the UNIVAC development and fund the payroll for their 134 employees in Philadelphia. So they approached IBM (where Mauchly behaved like a dick and would never fit in the IBM culture while Eckert was very neat) but because UNIVAC was one of IBM's few competitors, the antitrust law said they couldn't take them over. Eventually the entrepreneurs were so desperate that they were ready to consider the first reasonable offer: Remington Rand. The offer implied that Eckert and Mauchly now would become mere employees of the company they founded but they had no alternative, hence, they accepted. The UNIVAC operation was not disrupted or relocated and the giveaway orders of $150,000 were canceled because there was no way of making profit. As ever, Eckert was the hub around which the whole project revolved. Mauchly was no longer his companion but more

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the morale booster for everyone. The heat in Philadelphia made shorts and undershirts the uniform of the day. In 1951 the UNIVAC was completed and ready to commerce after some basic software was developed. The programming team was led by Grace Murray Hopper, who became the driving force behind advanced programming techniques for commercial computers and the world's foremost female computer professional. The following year more UNIVACs were completed and more orders were taken. In late 1952, Remington Rand achieved a spectacular publicity stunt for UNIVAC. The company persuaded the CBS television network to use UNIVAC to predict the outcome of the presidential election. A landslide victory for Eisenhower was predicted of 438 to 93 electoral votes, and the final result was indeed a win for Eisenhower of 442 to 89. Now UNIVAC was rapidly becoming the generic name for a computer. The appearance of the UNIVAC on election night was a pivotal moment in computer history. Before that date, while some people had heard about computers, very few had actually seen one; after it, the general public had been introduced to computers and had seen at least a mock-up of one. And that computer was called a UNIVAC, not an IBM.

IBM's big push

It was the launch of the Census Bureau UNIVAC eighteen months earlier that had galvanised IBM into action. They understood that they were losing business because they choose to develop the Defense Calculator (Model 701) rather than the MDC (Model 702) and TPM (Model 650). Hence, they decided to turn the MDC and TPM into a major push to counter UNIVAC. The development cost of the three concurrent projects was enormous, and large numbers of R&D workers had to be recruited. It also became clear that the rent for the Model 701 should be twice the initial amount of $8,000. Fortunately, most pre-existing orders held up at the revised price. This ready acceptance of the new price reflected the greater maturity and knowledge of computer buyers in 1952: large, fast scientific computers were going to cost about $1 million, regardless of who made them. Another maturation was that cost-effectives (which Model 650 lacked) was no longer the most important reason for buying a computer. Computers were hot news, like a 1952 article in *Fortune* magazine titled "Office Robots" caught. Also, IBM then realised that they had to be seen as an innovator and had to develop data-processing computers, whether or not it made money. When the Model 702 came into production, IBM had taken fifty orders, the actual deliveries took a while but because Remington Rand was unable to deliver more UNIVACs, IBM was enabled to protect its order book. Although the IBM 702 Electronic Data Processing Machine (EDPM) had superficially much the same specification as the UNIVAC, in the underlying engineering there were major differences that enabled IBM's product quickly to outdistance the UNIVAC in the marketplace. This was due to the Williams Tube memory that was more reliable and twice as fast. In addition, IBM's computers were modular (that fit into standard elevators), as opposed to the UNIVAC, this made shipment easier and led to greater flexibility in size and specification of the machine.

Finally, IBM's greatest advantage was its reputation as a service-oriented vendor. Although, neither the 701 nor the 702 proved to be quite the "UNIVAC-beater" that IBM had hoped, because they were less reliable than IBM's normal high standards. 1955, however, was the turning point for IBM because their 700 series computers increasingly exceeded those for UNIVACs. Yet, it was not this that secured IBM's leadership of the industry, but the low-cost MDC (it was actually expensive compared to machines from other makers) due to its superior engineering, reliability, and software, yielding revenues far exceeding those of the entire 700 series. IBM's understanding for marketing was also superior: they placed 650s in universities with discount so that a generation of computer scientists was created that nurtured on IBM 650s. This marketing mastery was more important than the technology mastery. With this they began to eclipse Remington Rand's UNIVAC division. Remington Rand's failure to invest heavily enough in computer software and marketing and the hassle of factories at different locations are other reasons for their decline. While UNIVAC was still the public's idea of a computer, corporate decision makers were beginning to favour IBM.

The computer race

By the end of the decade, as the computer was transformed into a business machine sold in high volumes, the electronics and control manufacturer companies were all faced with the same decision: whether to exit the business of to spend the enormous sums necessary to develop

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peripherals and applications software and pay for the marketing costs to compete with IBM. By far, the most vulnerable firms were the small computer start-ups, of which very few survived into the 1960s. Finding a niche in the field helped them retain their independence. The office-machine firms had, just like IBM, doubts about the market for business computers and initially did nothing more than put electronics into their existing products in an evolutionary way. NCR enhanced one of its ordinary accounting machines to "Post-Tronic," which was a brilliant success—bigger than almost any computer of its era. This made heavy investment for computer development by NCR possible.

Burroughs was very courageous in spending several millions of dollars on electronics R&D. But profits were low and he failed in developing a successful data-processing computer. Hence, he acquired the Electrodata Company (a start-up) for $20.5 million, an enormous sum at the time, but with their successful business computer and his sales force and customer base, it was soon able to secure an acceptable position in the computer company league. Several other office-machine companies tried to enter the computer race in the 1950s (by either internal development or acquisition), but none survived. Thus, by the end of the 1950s the major players in the computer indus- try consisted of IBM and a handful of also-rans: Sperry Rand (former Remington Rand), Burroughs, NCR, RCA, Honeywell, GE, and CDC. Soon, journalists would call them IBM and the seven dwarfs. Namely, IBM was growing rapidly (think fivefold) and in the next five years after, IBM would come to utterly dominate the computer market. The engine that would power that growth was the new IBM 1401.

6 The maturing of the mainframe: The rise of IBM

The 1957 film *Desk Set* was the first to feature an electronic computer. In this film, Tracy and Hepburn spar and banter, and eventually fall in love, the imposing EMERAC (Electro-Magnetic Memory and Research Arithmetic Calculator, obvious play on the ENIAC and UNIVAC) is gradually revealed to be charmingly simple-minded and nonthreatening. The film was a triumph of product placement for IBM and was prominently acknowledged in the film's closing credits. And while the film accurately reflects contemporary concerns about the possibility of technologically driven unemployment, its ultimate conclusion about the computer is unrelentingly positive: computers were machines for eliminating drudgery, not human employees. The film was a remarkably insightful mirror of a key period in the history of modern computing because it contained accurate reflections of contemporary attitudes.

<u>The IBM 1401</u>
IBM had hired industrial designer Eliot Noyes to update the machines' old-fashioned look. The 1401 was not so much a computer as a computer *system*. Most of IBM's competitors were obsessed with designing CPUs and tended to neglect the system as a whole. Processors were the rage, but IBM recognised that customers were more interested in the solution to business problems than in the technical merits. As a result, IBM engineers were forced into taking a total-system view of computers: programming, field service, training, spare parts, et cetera. This was what most differentiated IBM from its competitors and brought success to the IBM 650 and soon the 1401.

Most of IBM's customers were still resisting the computer and continuing to cling to traditional accounting machines (e.g. punched-card machine) because 1) the cost is significantly larger without guarantee of a better job done by the machine, 2) less reliable because it was tube-based, 3) programming staff had to be hired to develop applications software, 4) peripherals derived from the (perfectly fine) traditional accounting machines had to be purchased. The new Model 1401 however would be cheaper, faster, and more reliable than the 650 due to the use of transistors for tubes and core memory for the magnetic drum. An important new peripheral under development was a high-speed printer, capable of 600 lines per minute. The programming problem was hard because IBM wanted to avoid businesses having to heavily invest in retraining punched-card-oriented employees or hiring a programmer. Thus, they offered a new programming system called Report Program Generator (RPG). This enabled the employees to start writing their own business applications using familiar notations and techniques with only a day or two of training. The success

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of RPG exceeded all expectations, and it became one of the most used programming systems in the world. IBM also freely released software for common business functions such as invoicing, stock control et cetera which only needed very little modification. They also developed entire program suites for the industries that it served most extensively for free because of their dominant position in the market.

The IBM 1401 was announced in 1959 and cost around the same price as a medium-sized punched-card installation. Their forecast having to deliver around one thousand systems was an spectacular underestimate, for twelve thousand systems were eventually produced. This was due to the addition of the new 1403 “chain” printer (600 lines/minute) that IBM supplied with the system. Regular printers at that time cost around 1/4 of the IBM 1401, but to achieve the same speed as the 1403 four of those regular printers were needed. That means that the stored-program computer was sort of included for free! Only problem of 1401’s popularity was that IBM needed to cope with the flood of discarded rental equipment. Furthermore, IBM stock is the standard example of sensational growth sustained one many years. Anyone who invested around $5,000 would be worth 500 times more in 1959. As more and more 1401s found their way into the nation’s offices in their powder-blue livery, IBM earned a sinister new name: Big Blue.

IBM and the seven dwarfs

Meanwhile, IBM’s success was creating a difficult environment for its (seven) competitors. Regarding Sperry Rand, despite being the pioneer of the industry, it had never made a profit in computers (just broke even in 1963) and was gaining a reputation bordering on derision. Even when Sperry Rand brought in an aggressive new manager, Rader, the guy was forced to admit: “It doesn't do much good to build a better mousetrap if the other guy selling mousetraps has five times as many salesmen.”

Burroughs and NCR only had a market share of around 3 percent (less than Sperry Rand), and both their strategies were safety first: protect their existing customer base, predominantly in the banking and retail sectors, respectively. CDC (Control Data Corporation) was the fastest-rising star in the computer scene in 1960. It was only three years old but had already evolved a successful strategy by manufacturing mainframes with a better price performance than those of IBM. They focused on the markets where IBM hadn't much impact, like the scientific market. By 1963 CDC had risen to third place in the industry, not far behind UNIVAC (Sperry Rand). RCA, Honeywell, and GE (General Electric) were all powerful and big, hence, possible to compete with IBM, but the computer was not a core business but merely an attempt to enter a new and unfamiliar market. All these companies targeted IBM’s customers by launching models that were able to run the same software of IBM, Honeywell in particular. IBM was sufficiently nervous about its vulnerability to antitrust litigation.

Revolution, not evolution: System/360

Although in marketing terms the IBM 1401 had been an outstanding success, inside IBM there was a mishmash of incompatible product lines that threatened its dominance of the industry. Many IBM insiders felt that this problem could be re- solved only by producing a “compatible” range of computers, all having the same architecture and running the same software. All this has to be compared with IBM’s punched-card products, where a single range of machines (the 400 series accounting machines) satisfied all its customers. The resulting rationalisation of production processes and standardisation of components had reduced manufacturing costs to such an extent that IBM had no effective competition in punched-card machines at all. The biggest problem however, was not in hardware but in software. Namely, m (# of different computer models) × n (# of software packages) had to be developed and supported. Just as problematic were the software written by IBM’s customers because the computers were so narrowly targeted at a specific market niche, horrendous organisational disruption was caused during the changeover period. All these factors suggested that IBM would sooner rather than later have to embrace the compatible-family concept. Downsides were: 1) it was not clear if compatibility to such extent was technically feasible, 2) possible that it costs too much to remain competitive in the market, and 3) not everyone at IBM wanted to stop consolidating its current success with existing machines. But it

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was the software problem that was to determine product strategy, and by late 1960 it became necessary to go for the radical solution. The SPREAD (Systems, Programming, Review, Engineering, And Development) task group was established for planning purposes and to come to agreements more easily. At the end of the year the SPREAD Report stated that the creation of a New Product Line (NPL) replacing all existing computers was recommended. There was little enthusiasm for NPL because of the enormous costs. But carrying on in the old ways is even more dangerous. Considerable emphasis was placed on commercial secrecy. Projects had misleading code numbers for example that gave no hint of a unified product line. NPL was one of the largest civilian R&D projects ever undertaken. It was hard that it took place in different locations, hence, extensive (and expensive) telecommunications facilities were used to keep the development groups coordinated. In total everything cost around $5 billion.

The name System/360 suggested betokening all points of the compass and thus the universal applicability of the machines. The machines were announced one at a time over a period of a couple of years to ensure that existing orders wouldn't be canceled and this transition was how IBM managed things before. But when Honeywell launched the 200 and took more orders in one week than in their 8 years existence before the announcement, there was a noticeable falloff in IBM 1401 orders. For this reason, Watson decided three weeks later to launch the complete NPL all at once. The new System/360 was intended to obsolete virtually all other existing computers. Though risky, this business judgement paid off handsomely. The thousands of orders far exceeded IBM's ability to deliver so it took over three years. Meanwhile, IBM's employee head count increased by 50 percent, taking it to nearly a quarter of a million workers. System/360 has been called "the computer that IBM made, that made IBM." It was to be the engine of growth for the next thirty years.

The dwarfs fight back

Although the compatible-range concept was already being explored by most manufacturers, IBM's announcement effectively forced this issue. It was not impossible to still compete with the computer giant by simply producing better products. Namely, there were some serious flaws in its software and it did not support time-sharing, which enabled a machine to be use simultaneously by many users. But as always, technology was secondary to marketing in IBM's success. Hence the seven-dwarfs competitors aim at one of IBM's technical weak spots. RCA competed with IBM by copying their System/360 but with a 10 to 15 percent advantage in price/performance. Every enhancement had to be kept up with. Eight months later RCA launched the Spectra 70.

Honeywell aimed for product differentiation that is not compatible with System/360 (which is less risky). Taking their highly successful model 200 and turned it into model 120 and four bigger ones (1200, …). This comfortably brought them the desired 10 percent of IBM's existing 1401 customer base. Burroughs did something similar with its 500 series in 1966 as well as NCR with its Century Series in 1968.

Another strategy to compete with IBM was to aim for niche markets that were not well satisfied by IBM and in which the supplier had some particular competitive advantage. CDC, for example, recognised that IBM didn't offer really big machines, hence, CDC did. Primarily for government and defence research agencies. And GE offered, in collaboration with computer scientists at Dartmouth College and MIT, computers that allowed for time-sharing, which quickly became the world leader in time-shared computer systems. UNIVAC continued to compete successfully with IBM in airline reservation systems. With these strategies and tactics, all seven dwarfs survived into the 1970s—just. Because IBM appeared invincible, it had about 3/4 of the market for mainframe computers around the world. Therefore, some of the dwarfs were losing money heavily, so RCA and GE decided to give up the struggle and terminate their computer divisions. Their customer bases were bought up by Sperry Rand and Honeywell, respectively, which helped them gain market share at relatively little cost. This resulted in IBM and the BUNCH (Burroughs, UNIVAC (Sperry Rand), NCR, CDC, and Honeywell) rather than IBM and the seven dwarfs.

Systems men and management information systems

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To simply tell the story of the computer industry in terms of the major machine manufacturers—as if the mere invention of a new technology automatically ensures that it will be adopted—would be misleading. Computers were used by individual people, not faceless corporations. The adoption of new technologies is based on a complex calculus of risk, benefit, functionality, cost, and reliability—as well as personal, professional, and political considerations. Because computers became increasingly inexpensive in the 1960s, it was possible to apply computer power for various purposes. In many large corporations the routine administrative processes (EDP: electronic data processing) had already been delegated to specialist data-processing departments.

The systems men were professional computer enthusiasts: they were the postwar incarnation of the "systematisers" of business methods that emerged in the 1880s. They developed systems of management that would eliminate waste and inefficiency at all levels of the organisation: the management information system (MIS, not "automation"). Computers and communications technologies now could be mobilised to create a comprehensive picture of the total information environment of a complex and distributed organisation without needing middle managers. In this respect, computerisation was as much ideological as it was about measurable efficiency. Achieving the MIS was not just technically ambitious but enormously expensive, therefore almost no one realised this goal. Nevertheless, the idea of the computer as a tool for managerial control would have a effect for the next several decades. Despite the fears shown in *Desk Set*, the 1960s was characterised by an almost universal optimism about the impending computer revolution. "There seems to be no limit to the possibilities that scientists see for computers."

The long march of computerisation

The rapid dissemination of electronic computing was enabled by innovation and the standardisation of the computer industry around IBM equipment. More important was the development of human capabilities—the desire and expertise necessary to integrate computer technology into existing business practices and organisations. Management consulting firms aggressively promoted computerisation as an aid to achieving the larger goal of managerial and operational efficiency. Thus, Ross Perot founded Electronic Data Systems (EDS), the first of many computer facilities firms, which would establish and operate for its clients an entire computer installation. This made it possible for organisations to use technology without having the expertise. Computerisation became a widespread phenomenon. Some industries like medicine were even fundamentally transformed by computer technology. And in the financial sector opportunities like the ATM arose. Perhaps the most profoundly revolutionary applications that were developed for the digital electronic computer had to do with simulation and the ability to make mathematical models less expensive and more accurate (came in handy for meteorology and the petroleum industry for example).

The computer becomes a commodity

By the beginning of the 1970s, the computer customers were increasingly savvy computer users, no longer dependent on computer manufacturers like IBM to provide technical expertise or advice on systems integration. Regarding the System/360, its architectural standardisation eliminated certain forms of competition, but it introduced others as well. Companies could develop an entire alternative system of products that are completely interchangeable with their IBM equivalents, like Fujitsu did. Faced with this competition from lower-cost compatible competitors, the mainframe business suddenly became a lot less profitable. By the mid-1970s, the mainframe had matured and become a commodity. As a result, computer users from a diverse set of industries began to drift to other, less intimidating and less expensive suppliers than IBM. To be sure, IBM enjoyed an exceptional dominance of computing for at least two decades. But such dominance could not last forever, and it didn't.

PART III INNOVATION AND EXPANSION

7 Real time: Reaping the Whirlwind

While computers were more glamorous than punched-card machines, they were not necessarily

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more efficient or cheaper. Where computers *did* make a difference was in real-time systems. A real-time system is one that responds to external messages in “real time,” which usually means within seconds. No previous office technology had been able to achieve this speed, and the emergence of real-time computers had the potential for transforming business practice.

Project Whirlwind

The technology of real-time computing was first developed for a military purpose in MIT’s Project Whirlwind—a project that, in turn, grow out of contract to design an “aircraft trainer” during World War II. This would allow to train pilots faster by using a simulator (“trainer”). The electromechanical control system fed the appropriate data tot the aircraft’s instruments, and a series of mechanical actuators simulated the movement of an aircraft. A limitation was that a different simulator had to be built for each type of aircraft. Project Whirlwind comprised developing a universal flight trainer. The young Jay Forrester was placed in charge of the project, and although he was very resilient, the Whirlwind exceeded its original cost (forty times more) and time (quadrupled) estimates the most. However, the project did manage to survive and had an impact that eventually did justify the expenditure because it established real-time computing. It appeared to be troublesome to develop an analog computer that was fast enough to operate the trainer in real time. Perry Crawford had a significant impact in the course that Project Whirlwind was set in, because he was the one who suggested a general-purpose digital computer would potentially be faster and more flexible than an analog computer. Even though this would result in a major cost escalation, he believed that a general-purpose computer would be the solution to many scientific and engineering problems other than those associated with aircraft flight. Forrester staffed the project with young graduate engineers and gave them considerable independence. He wrote a proposal to the Special Devices Devision of the Bureau of Aeronautics (SDDBA) requesting the revised project (tripling the original budget) and succeeded. The single most difficult problem was the creation of a reliable storage technology since the mercury delay-line memory was far too slow, so he immersed himself in electrostatic storage tube. Years of fruitful developing passed by, but the Office of Naval Research (ONR, which took over SDDBA) had shrunk the defence budgets dramatically. Fortunately Forrester was blessed with one key ally in the ONR: Crawford, whose vision for real-time computing was undiminished. There were two reasons for Whirlwind’s much higher costs. First, it needed to be an order of magnitude faster than any other computer to fulfil its real-time function. Second, it needed to be engineered for reliability because breakdowns that would cause no more than a minor irritation in a routine mathematical computation were unacceptable in a real-time environment. When Crawford moved to another assignment, Forrester faced heavy weather with ONR. But he was persistent, he argued that Whirlwind was a national computer development of such self-evident importance that the costs, however great, were immaterial. This won ONR over for a little while, but unfortunately the electrostatic storage tubes had proved expensive and its performance was disappointing. But Forrester had an idea and worked on it by himself for several months. Then, ONR gave Forrester a last chance to bring Whirlwind to a working state. Fortunately, as one door shut another had already begun to open.

The SAGE defence system

America had to improve their defence system when it was revealed that the Russians were planning on dropping a nuclear bomb on the United States in 1949. The existing defence system dated back from WWII. The weakness of this system was its limited ability to process and make operational use of the information it gathered. There were communication difficulties at all levels of the system and gaps in radar coverage. George Valley recommended serious upgrading to fix these problems, this on behalf of his committee, the Air Defense System Engineering Committee (ADSEC or the Valley Committee). He ended up working together with Forrester from Project Whirlwind because he was very impressed by their machine. He even allocated $500,000 to Project Whirlwind. This transformed the perception of Forrester’s group within the ONR, and the criticism began to disappear. So, more money was invested to continue with the project and later the Whirlwind became the major component of a much larger project, Project Lincoln: the research and development program for a computerised national air-defence system. It would take another decade for its result—the Semi-Automatic Ground Environment (SAGE)—to become fully operational. But in 1951 Whirlwind already became operational and contained a core memory

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(developed by MIT and rather than the unreliable electrostatic memory) with an access time of nine microseconds—making Whirlwind by far the fastest and most reliable computer in the world. In five years time, core memory would replace all other computer memory types. This income would have justified the cost of the completed Whirlwind project. Gradually the technology was transferred to IBM, which transformed the prototype Whirlwind into the IBM AN/FSQ-7 computer, the largest computer ever put in service. In the Direction Centers (where air surveillance took place) this computer was duplexed for reliability: one machine operating live and the other in standby mode. The surveillants used CRT-based graphical monitors for this, which created a culture of interactive computing years later. Eventually, the SAGE system was constructed consisting of a network of Direction Centers, collectively deciding if overflying aircrafts were civilian or military and friend or foe by analysing their positions and velocities. Because of new developments in military defence, bombers became a second-order threat, which made SAGE's contribution to military defence marginal. However, its contribution was notably a technological by-product to civilian computing. Many of the technological innovations (key enabling technologies for the commercial exploitation of computers!) that were developed for SAGE were quick diffused throughout the computer industry. The SAGE system also provided a training ground for software engineering talent since some 1,800 programmer years of effort went into its software. IBM was undoubtedly the major beneficiary of the system. Watson Jr. said: "It was the Cold War that helped IBM make itself the king of the computer business, " because the Cold War tension provided the catalyst to get the nation's air-defence system reviewed. They obtained a lead in processor technology, mass-storage devices, and real-time systems, which it never lost and soon put to commercial use.

SABRE: A revolution in airline reservations

The first major civilian real-time project to draw directly on IBM's know-how and the technology derived from the SAGE project was the SABRE airline reservations system developed for American Airlines. SABRE, and systems like it, transformed airline operations and helped create the modern airline as we know it. Before computers, reservations were handled the following: reservations clerks would answer incoming calls regarding flight availability, cancelations, and bookings from customers or travel agents. The requests were referred to displayed availability boards, and for flights further ahead in time, the clerks had to walk across the room for card files. If the request was possible, its details would be recorded on a card which would be collected a few minutes later to update the availability boards. Connecting flights caused even more trouble and a day or two before the departure of flights the clerks had loads of work from unconfirmed passengers, last-minute bookings and cancellations. Tens of thousands of reservations were processed in this manner every day; in reservations offices at several different airports, there was almost no mechanisation. They operated in batch-processing mode: 1) batched in hundreds or thousands, 2) sorted, 3) processed, 4) checked, 5) dispatched. Pro was that this reduced the cost of each transaction, it was also cost-effective for other data-processing problems of businesses, but the con was that it could not address the instantaneous processing needs of an airline reservation system. Manually processing these reservations weren't a problem until the 1950s when air travel was considered a luxury purchase. But the airline market kept growing, thus in 1944 American Airlines let Teleregister develop Reservisor for them, which determines the availability using a specially designed terminal, eliminating the need for the availability boards. Its pro was coping two hundred extra flights in the first year with twenty fewer clerks. Its con was that the Reservisor was not directly integrated into the overall information system, so the sales still had to be recorded manually. For this reason about one in twelve reservations was in error. Therefore, most business travellers returning home on a Friday would have their secretaries book at least two return flights, which led to underselling a flight and financial loss, or overbooking a flights, with consequent damage to customer relations.

By 1953 American Airlines had reached a crisis. Increased traffic and scheduling complexity were making reservations costs insupportable. But they had to respond to increasing competition and the most effective way was by improving load factors—the proportion of seats occupied by fare-paying passengers. Hence, they bought thirty expensive Boeing 707s, which worsened this crisis. So when the president of American Airlines fortuitously met a senior salesperson from IBM in 1950, Watson Jr. was given a heads up about their crisis and he decided to enter the airlines

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reservations market as well as similar “tough applications e.g. Banks, Departments Stores, Railroads, etcetera.” They hired Crawford from ONR to lead real-time developments capitalising on IBM’s SAGE experience. While a computerised reservations system was a desirable goal in the long run, it would be some years before cost-effective technology became available. Before the computerised system would be into operation in 1959, an impressive, though soon-to-be-obsolete, new punched-card system and improved Reservisor were brought into operation. This later became known as a “technology intercept” strategy. In 1960 the reservations system was named SABRE, inspired by the Buick LeSabre and an acronym that was dropped later. SABRE or Saber suggested speed and accuracy in its name. The costs were around $40 million (same as five or six Boeing 707s). It was fully implemented during 1960-1963. The project was easily the largest civilian computerisation task ever undertaken up to that time, involving some two hundred technical personnel producing a million lines of program code. The central reservations system was based on two IBM 7090 mainframe computers (again duplexed for reliable operation) and sixteen disk storage units with a total capacity of 800 million characters, the larges online storage system ever assembled up to that time. Now, most transactions were handled in three seconds. More important than the volume of transactions, however, was the dramatic improvement in the reservations service provided by the system. Availability information was constantly updated and instantly accessible, containing information about the passengers e.g. including meal requirements and hotel. The system quickly took over not just reservations but the airline’s total operation: flight planning, maintenance reporting, crew scheduling, fuel management, air freight, and most other aspects of its day-to-day operations. It was fully operational in 1964 (took ten years to implement) and amply recouping its investment through improved load factors and customer service. And although it was known as “the Kids’ SAGE” it was a massive project in civilian terms and not without risk. What was innovation for one airline soon became a competitive necessity for the others. All major carriers possessed reliable real-time systems by the early 1970s. A hint of the extraordinary developments that followed: by 1987 United Airlines used eight powerful IBMs to process over one thousand internal and external messages in a single second. Eventually, the airline reservations problem was unusual in that it was largely unautomated in the 1950s and so there was a strong motivation for adopting the new real-time technology. Longer-established and more traditional businesses were much slower to respond. But where the airlines pioneered, traditional businesses eventually followed (like ATMs).

The Visa system and Automated Teller Machines

There were two roots to the financial revolutions involving credit cards and ATMs: first, the idea of “cashless” payment cards and, second, the development of cash- dispensing machines. Before the ATM, the Diners Club was the first independent payment-card organisation, a cardboard booklet could be used to pay for whatever, which the cardholders were billed for monthly. There was also already a cash-dispensing machine De La Rue Automatic Cash System (DACS) into operation by Barclays in North London, they were the first in the world in 1967. The customers could purchase punch-card vouchers (“Barclaycash”) to be inserted in the machine, along with a six digit authorisation number, to obtain cash. Part of the card was cut off and remained in the machine for a bank employee to later complete the transaction. De La Rue started working for NCR, which, along with IBM, moved into the market. Chemical Bank eventually became the first bank to deploy an automated cash dispenser in the United States using the ubiquitous magnetic strip containing the card’s basic information (an invention from IBM from nearly a decade earlier). Their subsequent model in 1971 became a true ATM by allowing customers to make deposits, check balances, and transfer funds between savings and checking accounts, transiting the society to being check less. This resulted in a reduction in paper handling and a decrease in transaction times, errors, and costs, but required considerable technological, organisational, and cultural change. With payment cards, the artefact is quite basic whereas the vast and complex underlying technological system is largely invisible—the real-time computers. The Visa payment system became the largest of the world. Motivation for payment cards was the desire to compete for profits in an otherwise highly regulated industry as well as by the growing check-processing burden. Bank of America introduced the BankAmericard “credit card” in 1958 was soon profitable and had a virtual monopoly, but had the critical problem of a customer’s credit limit, and purchases above the limit needed authorisation via telephone approval. Slow settlement and clearing led to fraud. Other

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banks decided to establish their own cooperative issuer called Visa. Visa was led by Dee Hock, a commanding and eccentric person who had extraordinary visions that are still relevant today e.g. any account should be available for electronic-based purchases or transfers worldwide at any time. But authorisation of purchases remained a problem. Hence, Hock decided to use real-time computers and two computer systems—BASE I (authorisation within seconds) and BASE II (clearing and settlement)—were built. The former was delivered on schedule and within budget. For the latter, Hock sought an automated computer system that eliminated the mailing of paper by transforming the sales drafts into electronic records and clearing them through a BASE II centralised computer. It could process over ten times faster, resulting in far quicker fraud detection and saved labor and postage costs of over $15 million in its first year. This for a system that stayed within its $7 million development budget en was delivered on time in 1974. In the early 1980s Visa incorporated magnetic stripe technology and inexpensive point-of-sale terminals that connected directly with BASE I. With these state-of-the-art systems in place, the number of accepting merchants worldwide, member institutions, and issued Visa cards all more than doubled between 1975 and 1983, while sales volume went up more than five-fold to $70 billion. In accord with Hock's grand electronic-value-exchange vision, Visa also introduced an "asset card," initially named Entrée, in 1975. Drawing directly on a cardholder's bank account, it was soon renamed Visa Debit. Adoption was slow at first, but soon became a common form of payment, subsequently, these cards were linked to ATM networks. Like payment cards, ATMs were a transformative technology in banking, allowing automation for many simple transactions that replaced time-consuming processes for bankers. Adoption was slow here too, but in a decade also smaller banks had ATMs and national networks were launched so that customers were not limited to using only their own bank's ATMs. Hock's electronic value exchange vision was realised with online banking in the 1990s — the World Wide Web era.

The Universal Product Code

The Universal Product Code (UPC) is the barcode now almost universally found on food products manufactured or packaged for retail. The most important early innovation of the supermarket was the labor-saving concept of moving customers past the goods in a store (that is, self-service) instead of having a salesperson bring the goods to the customer. The recession in the early 1930s provided the right economic conditions for the cheaper supermarkets like A&P to thrive in Eastern United States, huge structural changes were afoot here. Their traditional grocery stores were replaced by supermarkets, so by the end of the 1960s the supermarkets accounted for three-quarters of the total grocery business. Because of the productivity and high inflation in 1970, supermarkets had to reduce costs and the most obvious area for improvement was the checkout. Hence, plans were made to develop terminals that offered the possibility of automatic data capture for pricing and stock control. But product codes were not generally feasible for supermarkets: the large number of product lines offered (10,000 to 20,000) and the low average value of items made the individual application of product codes too expensive. Several attempts to implement product coding were operationally successful but not cost-effective. Product coding would only be economic if individual fits acted cooperatively, or even better if the manufacturers, wholesalers, and retailers all used the same universal product code to become one integrated operation. This implied an extraordinary degree of cooperation among all these organisations (all-or-nothing basis), and the implementation of the UPC was thus as much a political achievement as a technical one. McKinsey & Company acted as consultants and opted for a top-down approach. This would allow a consensus to be reached that was acceptable to all. Simultaneously, many technical decisions had to be made like 1) the ten digits code (five representing the manufacturer, five representing the product. Ultimately a bad decision, Europe adopted 12 digits code), 2) the product code would be applied by the manufacturer, because it's cheaper than in-store application by retailers, and 3) the article number would not contain any price information because prices differ from store to store.

By mid-1971 the principle of a UPC had been accepted throughout the industry, but agreeing on the technology was also hard. Applying the code to products had to be inexpensive in order not to disadvantage small manufacturers; and the expense of product coding could not add significantly to the cost of goods, as this would disadvantage retailers who were unable to participate in the

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system because they would be forced to bear the increased cost of the system without deriving any benefit. The checkout equipment also had to be relatively inexpensive because millions of barcode readers would eventually be needed. This system was largely developed by IBM because it was the cheapest and most trouble-free option. The scanning technology was inexpensive and the barcode could be printed in ordinary ink. The first deliveries of barcoded merchandise began in 1974. Because of the enormous re-equipment costs involved, the take-up of the new technology was much slower (mid-1980s) among food retailers than it had been among food manufacturers, who simply had to redesign their labels. The barcode principle diffused very rapidly to cover not just food items but most packaged retail goods, it had become an icon of great economic significance. It was also a big pro that shelves did not have to be empty before the restocking process is started because of the added information flow among manufacturers, distributors, and retailers. While it is true that barcoding caused checkout and inventory costs to be reduced, this was not the primary reason for a renaissance of supermarket profits, which rose to an all-time high in the 1980s. A more important source of profit was a greatly increased product range—in some cases the number of lines more than tripling—as a result of checkout automation. Many of these new lines were luxury goods with large margins; these were the real generators of profits. The success also illustrates the political effect of trust in real-time computing. There were also negative impacts of bar coding, like the barriers to entry into food manufacturing became greater. The manufacturer now needed an operation of substantial scale to penetrate the retail network, even ironically the “folksy specialty foods.”

8 Software

Completing a machine was only the first step in the larger process of developing a working computer system, software was a significant element as well and this wasn't easy. The costs and challenges associated with developing computer programs, or “software,” became increasingly apparent in the 1950s. Even as computer technology was becoming ever smaller, faster, and more reliable, computer software development projects acquired a reputation for being over-budget, behind-schedule, and bug-ridden.

<u>First steps in programming</u>
Inside every stored-program computer (including those of today) programs are stored in pure binary—a pattern of machine representations of one and zeros. But because humans cannot readily use long strings of binary digits, the notation was altered from “11100000000110010” to the more programmer-friendly “A 25 S” for the EDSAC. Von Neumann and Goldstine (from the EDVAC) assumed that the programmer had to converse the program into binary, but Wilkes realised that this conversion could be done by the computer itself. This insight is only obvious in retrospect. Wilkes turned over the programming problem to David Wheeler (a 21 year old Cambridge research student) and he wrote a thirty line program (“Initial Orders”) that would convert programs to binary. Writing a program called for a fresh outlook, unbound by convention, and a very high order of manipulative technique, thus a young person’s game. After Wheeler solved one problem, the second one had to be dealt with: actually getting programs to work correctly.

It was important to check programs before running it, since machine time cost, in the early 1950s, around $50 an hour. However, almost all newcomers to programming had to learn the truth about debugging in the painfully slow and expensive way. The Cambridge team developed a “subroutine library” that consisting of many common operations that one could copy in their program to reduce the number of errors. The name can be attributed to the fact that programs used to call routines. The idea of reusing existing code was and remains the single most important way of improving programmer productivity and program reliability. Wheeler’s solution to the library was simple and elegant. Each subroutine was written so that, when it was loaded into the memory immediately following the previous subroutine in the program, it would run correctly. The Cambridge team put all their programming ideas into a textbook entitled *The Preparation of Programs for an Electronic Digital Computer*, which Wilkes chose to publish in the United Stated to reach the widest possible readership. Generally, designers had given little thought to programming until they suddenly had a

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working machine. The Cambridge model thus set the programming style for the early 1950s and even today the organisation of subroutines still follows this model.

Flowcharts and programming languages

Programming a computer is, at its essence, a twofold series of translations: 1) the translation of a problem into an algorithm that can be performed by a machine; and, 2) the translation of this algorithm into the specific instructions (or language) used by a particular computer. Both types of translations were fraught with difficulties and the potential for error and miscommunication. The earliest tool developed to facilitate the former translation was the *flowchart*. Flowcharts were intended to serve as the blueprint for computer programmers, an intermediate step between the analysis of a system and its implementation of a software application.The latter had partially been addressed by Wilkes and Wheeler. However, in 1953 the centre of programming research shifted to the United States because there was a desire to exploit the increased potential of large memories by designing more sophisticated programming techniques than that sufficed for British computers with smaller memories. “Automatic programming” systems comprised writing a program in high-level programming code that looked like English or algebra (thus easily readable) and that computers could convert intro binary. In brief, the mechanical translation of a design flowchart was incorporated as much as possible. This was a technological and a cultural problem, because the craft is so new and many programmers were reluctant to change their working methods. Grace Murray Hopper (first programmer for the Harvard Mark I and UNIVAC) stimulated change of this conservative culture more than anyone. In 1954 the outstanding problem in programming technology was to develop an automatic programming system that produced programs as goos as those written by an experienced programmer. John Backus from IBM tackled this problem the best by producing the economic system Formula Translator (FORTRAN)—the first really successful “programming language.” Backus argued that “programming and debugging accounted for as much as three-quarters of the cost of operating a computer; and obviously, as computers got cheaper, this situation would get worse.” A majority of the people at IBM thought Backus was attempting the impossible. It took Backus 2.5 years to write eighteen thousand LOC in the FORTRAN system and was released in 1957. The result was a system that worked on any machine for which a translator was available and that could give explicit diagnostic statements regarding the program that users try to run. FORTRAN was a phenomenal aid to the productivity of a programmer. In 1959 FORTRAN II was developed from the feedback from the many users. Although IBM’s FORTRAN was the most successful scientific programming language, there were many parallel developments by other computer manufacturers and computer users around the same time, e.g., MAD (Michigan), IT (Carnegie-Mellon), ALGOL (Europe). However, within a couple of years FORTRAN had come to dominate all the other languages for scientific applications, because it simply was the first effective high-level language to achieve widespread use by introducing FORTRAN to educational places. The overwhelming attraction of FORTRAN for scientific applications was that it had become a “standard” language accepted by the whole user community. For business applications COBOL (COmmon Business Oriented Language) emerged in 1960. This language had to prevent having to rewrite all programs when businesses decide to change their computers. Its design was heavily influenced by the Flow-Matic data-processing language Hopper had developed at UNIVAC, which very much looked like English sentences. This provided a comforting illusion for administrators and managers that they could read and understand the programs their employees were writing, even if they could not write them themselves. While many technical people could see no point in such “syntactic sugar,” it had important cultural value in gaining the confidence of decision makers. Computer manufacturers initially only adopted their own languages, but when the government declared in late 1960s that it would not work with computers without a COBOL compiler unless its manufacturer could demonstrate that its performance would not be enhanced by the availability of COBOL. No manufacturer ever attempted such a proof, and they all immediately fell in line and produced COBOL compilers.

Software contractors

Although programming languages and other utilities helped computer users to develop their software more effectively, programming costs remained the single largest expense of running a

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computer installation. Many users preferred to minimise this cost by obtaining ready-made programs wherever possible, instead of writing their own. Recognising this preference, in the late 1950s computer manufacturers began to develop applications programs for specific industries such as insurance, banking, retailing, and manufacturing. There was no concept in the 1950s of software being a saleable item: most manufacturers saw applications programs simply as a way of selling hardware (with an increased price of course). Another source of software for computer users was the free exchange of programs within cooperative user groups (SHARE for users of IBM computers and USE for UNIVAC users) and offering libraries of customer-developed programs.

The high salaries of programmers, and the difficulty of recruiting and managing them, discouraged many computer users from developing their own software. This created the opportunity for the new industry of *software contracting*: firms that wrote programs to order for computer users of either large (governmental) corporations or modestly sized companies. The SAGE software developer RAND Corporation was the first major (huge!) firm to enter the sector of software contracting with a separate organisation called the System Development Corporation (SDC), afterwards, other major defence contractors followed. The SAGE software development has been described as a university for programmers. It was an effective, if unplanned, way to lay the foundations of the supremacy of the US software industry. The computer manufacturers also remained an important (yet, secondary) sector of the software contracting industry.

Computer Usage Company (CUC) was the first small-time software contractor founded in 1955 by two programmers from IBM. It was not expensive to set up in the software contracting business, because “all one needed was a coding pad and a pencil.” Namely, the client’s own machine could be used. When enough contracts had been successful and CUC had built up to a total of fifty-nine employees, they bought the firm’s first computer in 1959. A few years later they joined with other start-ups with similar growth patterns. The first half of the 1960s was a boom period for software contractors. By this time the speed and size of computers, as well as the machine population, had grown by at least an order of magnitude since the 1950s. This created a software-hungry environment. CUC for example had grown into a major firm with seven hundred staff members and an annual sale of $13 million.

<u>The software crisis</u>
Finding enough experienced programmers to do the work that was so desperately needed was hard. In the 1960s the industry observers even talked about a looming “software crisis” that threatened the future of the entire computer industry. This crisis impacted technological, economic, and managerial developments within electronic computer. One obvious explanation for the crisis was that the power and size of computers were growing much faster than the capability of software designers to exploit them. Where hardware performance improved of a hundred, software technology had advanced hardly at all. Projects involving MLOC frequently ended in disaster, thus, by the end of the 1960s, software development projects had become “a scare item for management… an unprofitable morass, costly and unending.” This especially applies to IBM’s OS/360 because all other mainframe manufacturers were supplying this “operating system” as well. Then when IBM began planning System/360, they also planned for software because it had become an essential complement to the hardware. For small to large computers they developed BOS (Batch …), TOS (Tape …), DOS (Disk …), and OS/360. The first three represented the state of the art in software technology and were completed within their expected time and resources, but the latter was very ambitious and became the most celebrated software disaster story of the 1960s. The problem with it was it being the most complex program artefact that had ever been attempted. All code had to work in concert without a hiccup. Frederick Brooks, who led the project, supposed that the way to build a very large program was to employ a large number of programmers. But that would not work with complex structures like software, so this fact lay at the heart of the software crisis. Incorporating “multiprogramming”—which would enable the computer to run several programs simultaneously—was an added difficulty, because it was unproved technology (although a marketing necessity). As schedules began to slip quickly—not for any specific reason, but for a myriad of tiny causes—more people were added to the development team. Early test trials showed that OS/360 was painfully slow and needed extensive rewriting.

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Then in 1965, fundamental design flaws emerged for which there appeared to be no easy remedy. For the first time, OS/360 “was in trouble from the standpoint of sheer technological feasibility.” In 1966 it was public knowledge that IBM’s OS/360 had software problems. The mood grew desperation and the only way forward seemed to assign yet more programmers (at the peak 1000+ programmers). This was later recognised by Brooks as being precisely the wrong thing to do. When OS/360 was released a year late, IBM had quadrupled its budget ($500,000,000), the single largest expenditure in company history, plus it took tool of people too; managers and programmers struggled with stresses ranging from technological to physical. But OS/360 merely being late was not the disaster part, when it was released, the software contained dozens of errors that took years to eradicate. Fixing one bug would often simply replace it with another. Lesson was: big software projects are expensive and unreliable.

Software as an industrial discipline

Although the existence of a software crisis was widely acknowledged, interpretations of its underlying nature and causes differed wildly. The most common explanation of the crisis was the idea that “computer people” represented a new and different breed of technical expert: it was regarded “black art” rather than a scientific discipline, because there was no formal academic program in programming and it was not clear which skills and abilities were needed. This made training and recruitment difficult. Existing programmers were self-taught, developed idiosyncratic techniques and solutions, and tended to work largely in isolation from a larger community of practitioners. Von Neumann’s assumption that programming was routine work that could be done by relatively unskilled (and low-paid, therefore often female) labour—soon proved inaccurate. The relative scarcity of skilled and experienced programmers, combined with a growing demand for programming labor, meant that most programmers in this period were well paid and highly valued. This created resentment from corporate employers whom felt that they were being held hostage by the difficult-to-manage, socially awkward, and “slightly neurotic” programmers: the stereotype of the computer nerd was well established in the industry literature and popular culture.

One solution attempted to the “programming problem” was to reduce the level of expertise required of programmers. IBM developed PL/1 (similar to COBOL) that was even more explicit in their claims to eliminate corporate dependence on expert programmers. But technological solutions were rarely successful. Another—more promising—approach to rationalising the process of programming was to develop a more rigorous academic approach to software development. The first professional society for computer specialists was founded in 1946 as the Association for Computing Machinery (ACM) that shifted its focus to software in 1960. In 1968 they issued its Curriculum ’68 guidelines, which helped standardise education in computer science programmes throughout the United States. The emergence of computer science programmes in universities helped solve some critical problems for the software industry: programming practices were improved and programming’s status raised to a legitimate intellectual discipline. But, this did not produce enough programmers to satisfy the demands of the larger commercial computing industry, because 1) programs took too long and excluded too many people, and 2) focus on theoretical concerns were not as relevant as the practical concerns to the applied work corporate programmers face. Hence, vocational trade schools in computer programming emerged as an alternative to university education.

The most visible and influential response to the software crisis was the “software engineering” movement sprang in Germany in 1968. They emphasised the need to impose industrial discipline on informal and idiosyncratic craft practices of programmers. Software developers ought to adopt methods and techniques borrowed from traditional manufacturing. The most successful programming idea was “structured design methodology,” where one would begin with an overview of the software artefact and when satisfactory move one level lower and so on. Techniques for this was developed in FORTRAN, COBOL and new languages like Pascal (1971). Another idea was to use formal methods—simplifying and mathematising the design process—, but technically they only worked well on relatively small programs and culturally they were intimidating for rusty math skilled software practitioners (too academic). The idea of a development model, which was as much a management tool as a technical one, was more successful. It recognised that the software

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writing process evolves in an organic way because of it life-cycle: as software becomes increasingly absolute in time, the process of conceiving, specifying and developing would begin again. In the end, software engineering proved to be only a partial solution to the problems associated with software development, because software production continued to prove resistant to industrialisation due to its hard to define and delineate nature; it was a difficult, multidisciplinary and organisationally disruptive undertaking.

Software products

In the late 1960s the process of hiring a software team for custom-produced software shifted to the purchase of a ready-made software product. Despite the improving software techniques that emerged toward the end of the 1960s, the gulf between the capabilities of computers to exploit software and its supply continued to widen. It was only economically feasible for the very largest corporation because of the huge development costs. For this reason, when the first stable (and standard) System/360 launched, software contractors started to develop software products—packaged programs—that would be much more cost-effective than custom software because of more sales for the same product. The first computer program sold as a product was Outflow by Applied Data Research (ADR), which generated the flowchart the corresponded to an already existing program. The development of the software-products industry was accelerated by IBM's unbundling decision of December 1968, by which it decided to price its hardware and software separately (prior to this, they sold a total-system with software for "free"), because they were under investigation for alleged monopolistic practices by the antitrust division of the Department of Justice. They still got sued for more than a decade until it was eventually dismissed. Within three years the software products market became fully established after IBM's unbundling that triggered an explosion of software firms and packages. Existing computer users like Boeing also started to sell their software products to recover their development costs. Perhaps the most spectacular beneficiary of the new environment for software products was Informatics, the developer of the top-selling Mark IV file management system—one of the first database products, which was already on the market before the unbundling. They recognised the niche and had the resources to develop a good product, which had modest success at first because computer users were not used to separate charges for software, but experienced explosive growth after the unbundling. It even had its own user group—called the IV (pronounced "ivy") League and was the world's most successful software product for fifteen years, until 1983, by which time it had cumulative sales of over $100 million.

After the unbundling, mid-1970s, all existing computer manufacturers were important in the software-products industry, but IBM remained a major player despite its mediocre products. This was because of 1) the sales to its huge software customer base and 2) its ability to lease its software to generate cash flow. The most successful software-products suppler after that was Computer Associates. They followed a strategy of growth by buying out other companies with the sole requirement of having "legitimate products with strong sales." Then firing half the staff of the companies it acquired. In fifteen years they took over more than two dozen software companies, including some of the very largest. By 1989 it had annual revenues of $1.3 billion and was the largest independent software company in the world. After that, their product line was eclipsed by Microsoft and other leading personal-computer software companies.

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